

THE UNIVERSITY OF DANANG
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FACULTY OF ADVANCED SCIENCE AND TECHNOLOGY



CAPSTONE PROJECT REPORT
M1 REPORT

**DESIGN AND IMPLEMENT OF AN
INTELLIGENT EDGE AIOT WEARABLE
SYSTEM FOR REAL-TIME SAFETY
MONITORING OF ELDERLY PEOPLE**

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1. Proposed methods

1.1. System Architecture Block Diagram

The system consists of three main modules operating in a layered architecture:

1. Data Acquisition Module: IMU sensor (BMI160) providing 3-axis accelerometer and 3-axis gyroscope data at 50 Hz, and PPG sensor (MAX30102) for heart rate and SpO2 measurement.
2. Edge Processing and Decision-Making Module: ESP32-S3 microcontroller (Xtensa LX7 dual-core, 512KB SRAM) running an on-device TinyCNN model for real-time fall detection with zero cloud dependency.
3. Communication and Alert Module: BLE 5.0 for low-power data synchronization with the companion mobile application, which triggers automated emergency calls/SMS upon fall detection.

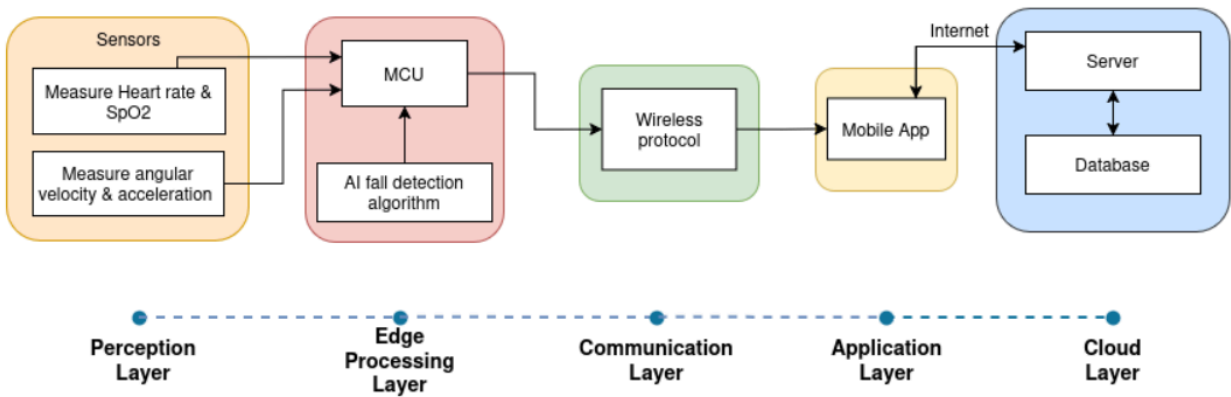


Figure 1: System Architecture Block Diagram

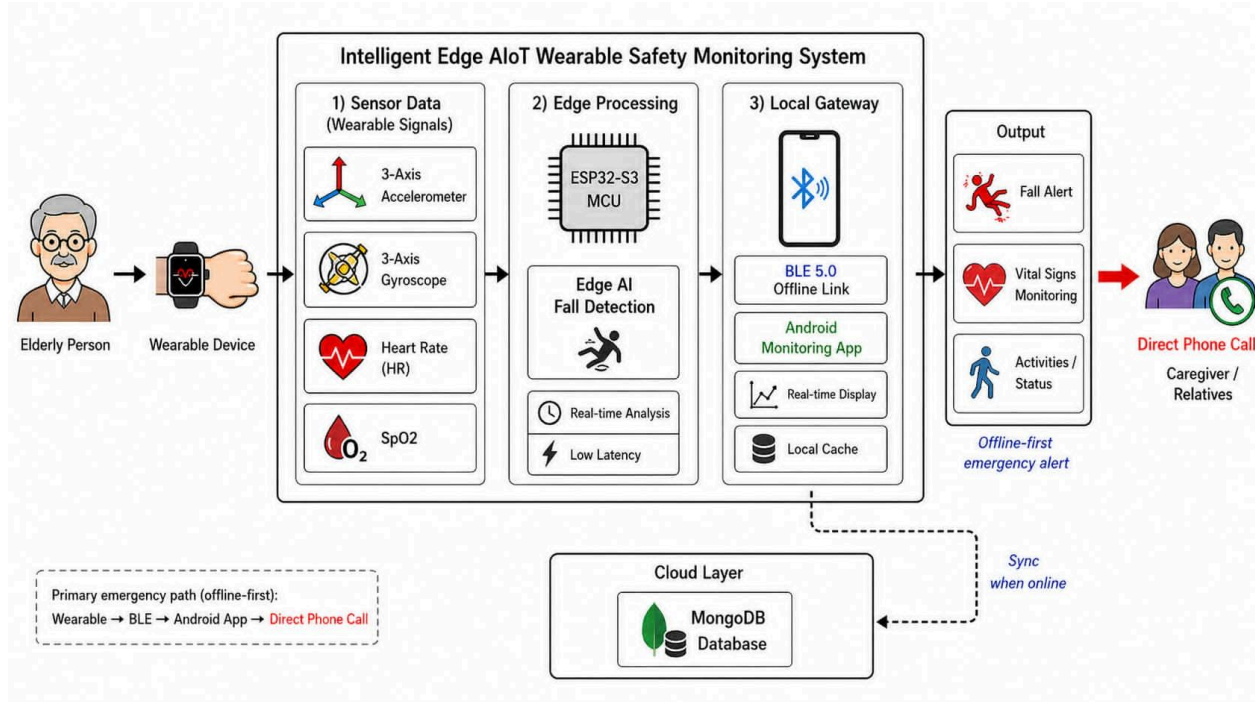


Figure 2: Data Flow Diagram

1.2 Hardware Component Selection

Components were selected through rigorous benchmarking (detailed in M1 report):

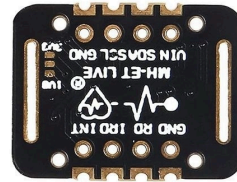
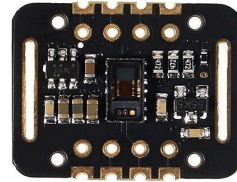
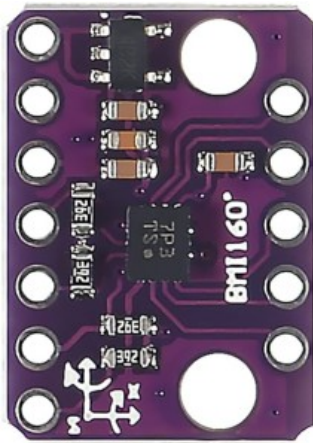
Table 1: Hardware Component Selection

Component	Selected	Key Reasons
Physiological Sensor	MAX30102	Integrated LED+PD+AFE, I2C, ~600uA, excellent wearable suitability
Motion Sensor (IMU)	BMI160	6-axis (accel+gyro), low noise (150ug/sqrt(Hz)), 950uA, compact 2.5x3x0.8mm
Microcontroller	ESP32-S3	Dual-core Xtensa LX7, 512KB SRAM, TFLite Micro support, BLE 5.0 + WiFi

Signal Processing Pipeline:

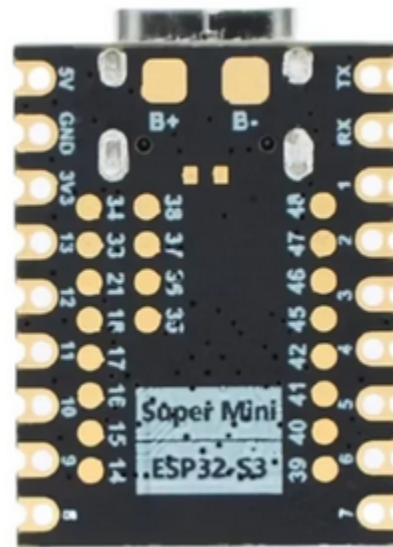
- IMU data: Raw 6-axis data -> Noise filtering -> Sliding window (100 samples x 6 channels) -> AI inference

- PPG data: Raw optical signal -> Signal conditioning -> HR and SpO2 feature extraction -> App display



(a) 6-axis Inertial Measurement Unit

(b) Heart Rate and SpO2 sensor



(c) Microcontroller

Figure 3: Selected Hardware Components

1.3 AI Fall Detection Model

1.3.1 TinyCNN Architecture

The AI model was designed specifically for ESP32-S3 constraints, using a lightweight 1D Convolutional Neural Network (TinyCNN) architecture:

Input (100x6) -> Normalization -> Conv1D(16, k=3, ReLU) -> MaxPool(2) -> Conv1D(32, k=3, ReLU) -> GlobalAvgPool -> Dense(32, ReLU) -> Dense(1, Sigmoid)

Table 2: TinyCNN AI Model Design Specifications

Design Parameter	Value
Model name	balanced_hr_imu_tinycnn
Input shape	100 timesteps x 6 channels (ax, ay, az, gx, gy, gz)
Sampling rate	50 Hz (2-second window)
Stride	50 samples (1-second hop)
Output	Fall probability (sigmoid, threshold = 0.40)
Model size (TFLite)	10.71 KB (far below 500 KB limit)
Quantization	Full INT8 (TFLite Micro optimized)

Key design decisions: (1) 1D Convolutions efficiently capture temporal motion patterns; (2) GlobalAveragePooling reduces parameter count compared to Flatten+Dense; (3) INT8 quantization enables fast inference on the resource-constrained MCU; (4) At 10.71 KB, the model is 50x smaller than the 500 KB requirement.

[Figure 4: TinyCNN Model Architecture Diagram]

1.4 Dataset and Training Pipeline

Data source: HR-IMU fall detection dataset with wrist-mounted accelerometer and gyroscope data.

Table 3: Data Distribution Before and After Balancing

Stage	Fall Windows	Non-Fall Windows	Total
Before balancing	1,628	5,468	7,096
After balancing (undersampling)	1,628	1,628	3,256

Data split (stratified):

- Train: 1,140 fall / 1,139 non-fall = 2,279 windows (70%)
- Validation: 244 fall / 244 non-fall = 488 windows (15%)
- Test: 244 fall / 245 non-fall = 489 windows (15%)

Training Configuration:

Table 4: AI Model Training Configuration

Parameter	Value
Optimizer	Adam (lr = 1e-3)
Loss function	Binary Cross-Entropy
Epochs	60 (Early Stopping, patience = 10)
Batch size	32
Best model selection	Checkpoint with lowest validation loss

1.5 Mobile Application

A companion Android application was developed to serve as the user/caregiver interface.

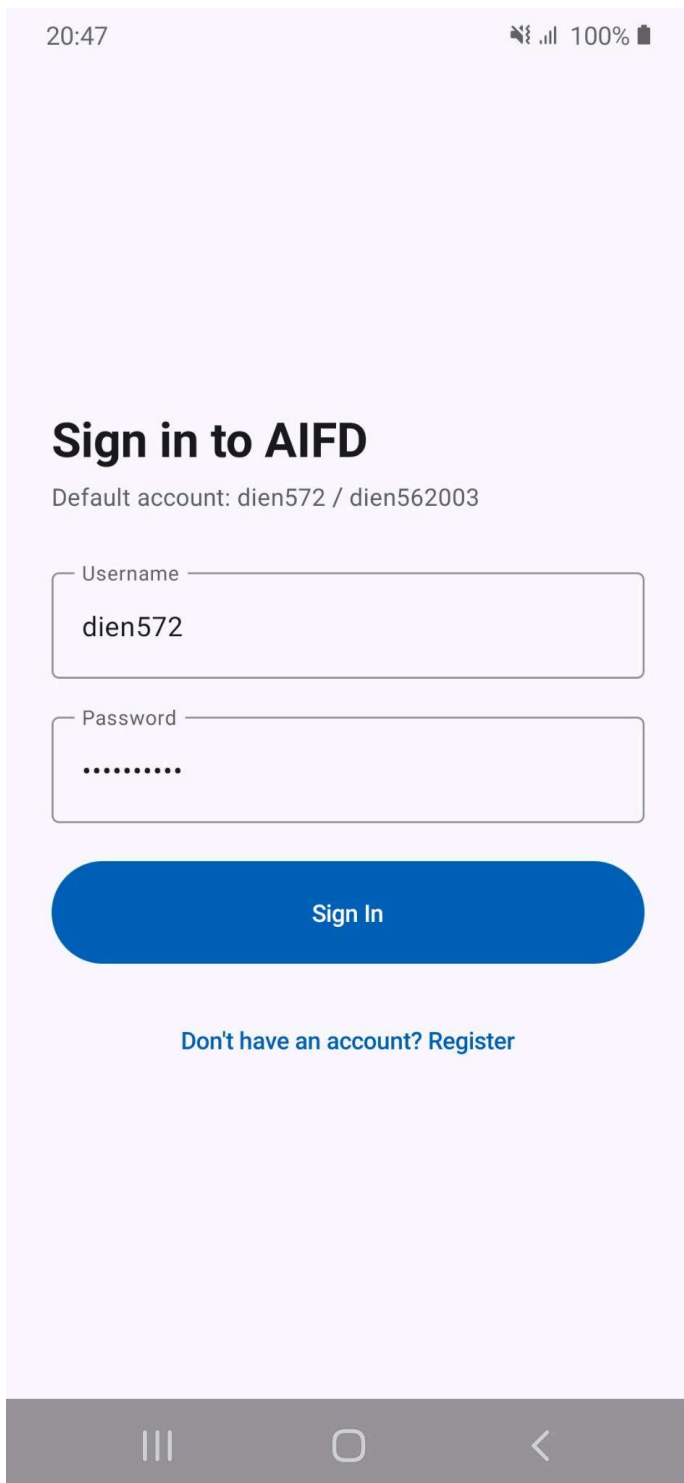
The app communicates with the wristband via BLE 5.0 and provides:

- Live HR and SpO2 monitoring display
- Historical health data visualization and trends
- Fall detection alert notifications

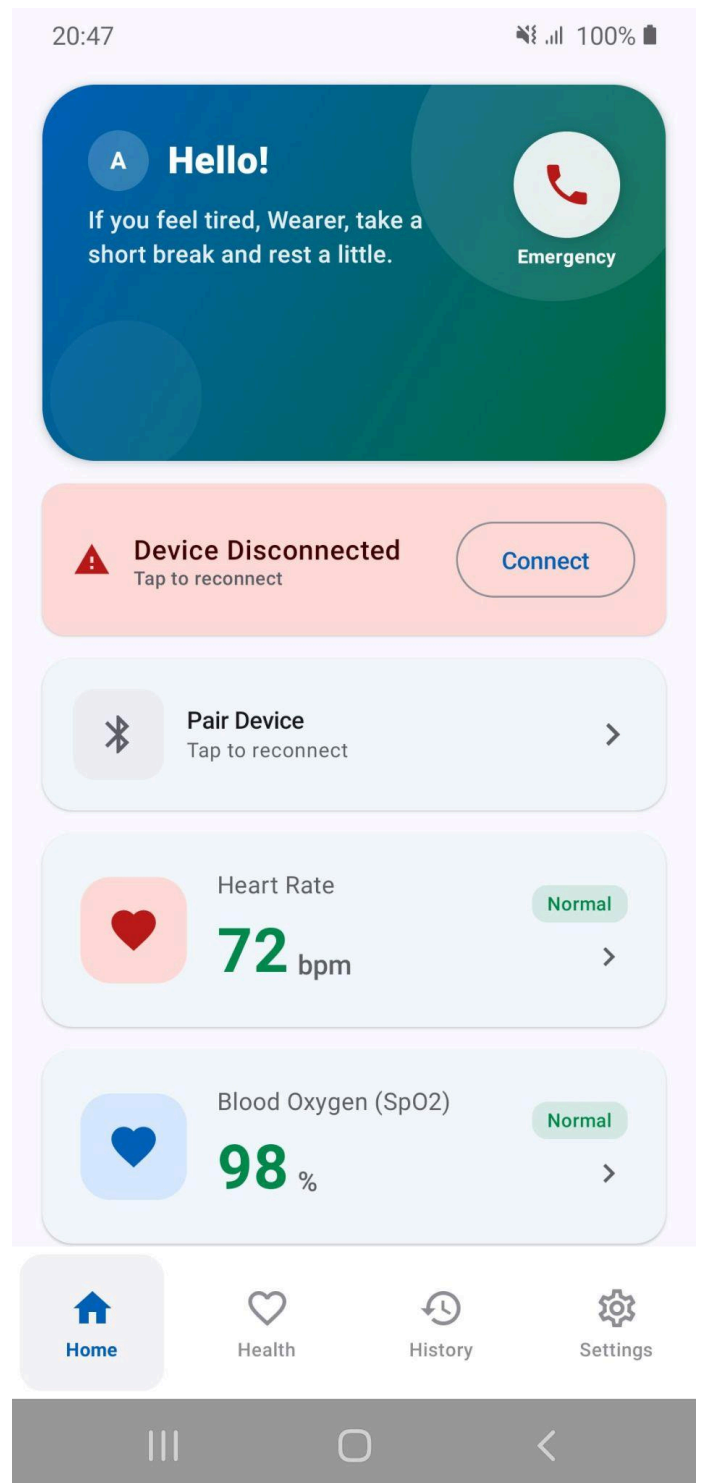
- Emergency contact management (auto-call/SMS configuration)
- BLE 5.0 connection management with automatic reconnection



Figure 5: App Icon

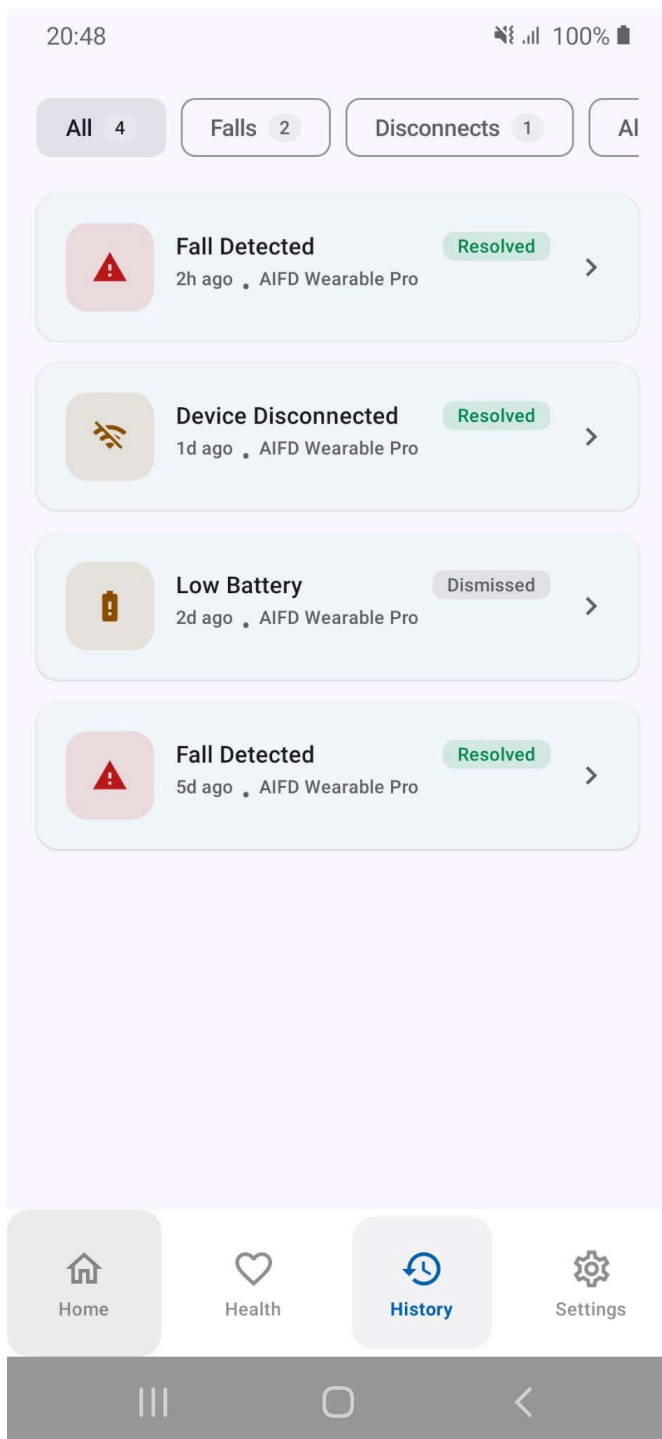


(a) : Log in interface

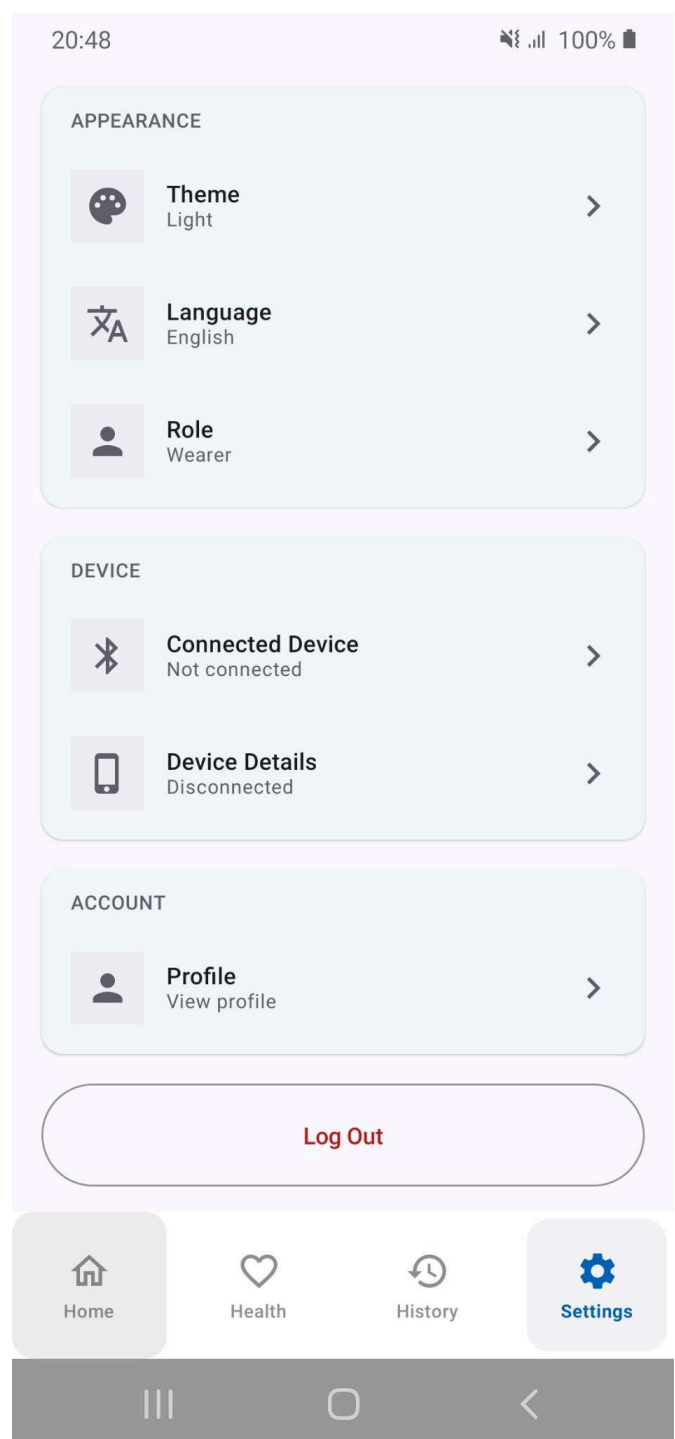


(b):Home interface

Figure 6:



(a) : Home interface



(b) : Setting interface

Figure 7:

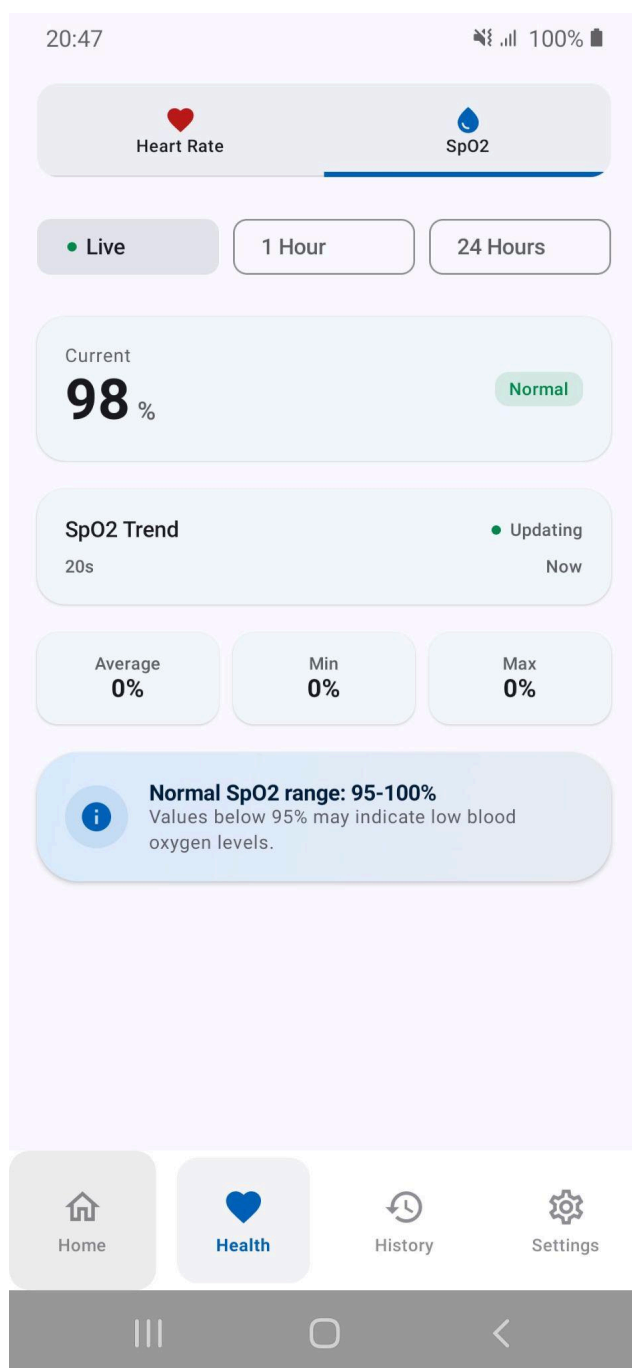


Figure 8: Heart rate interface

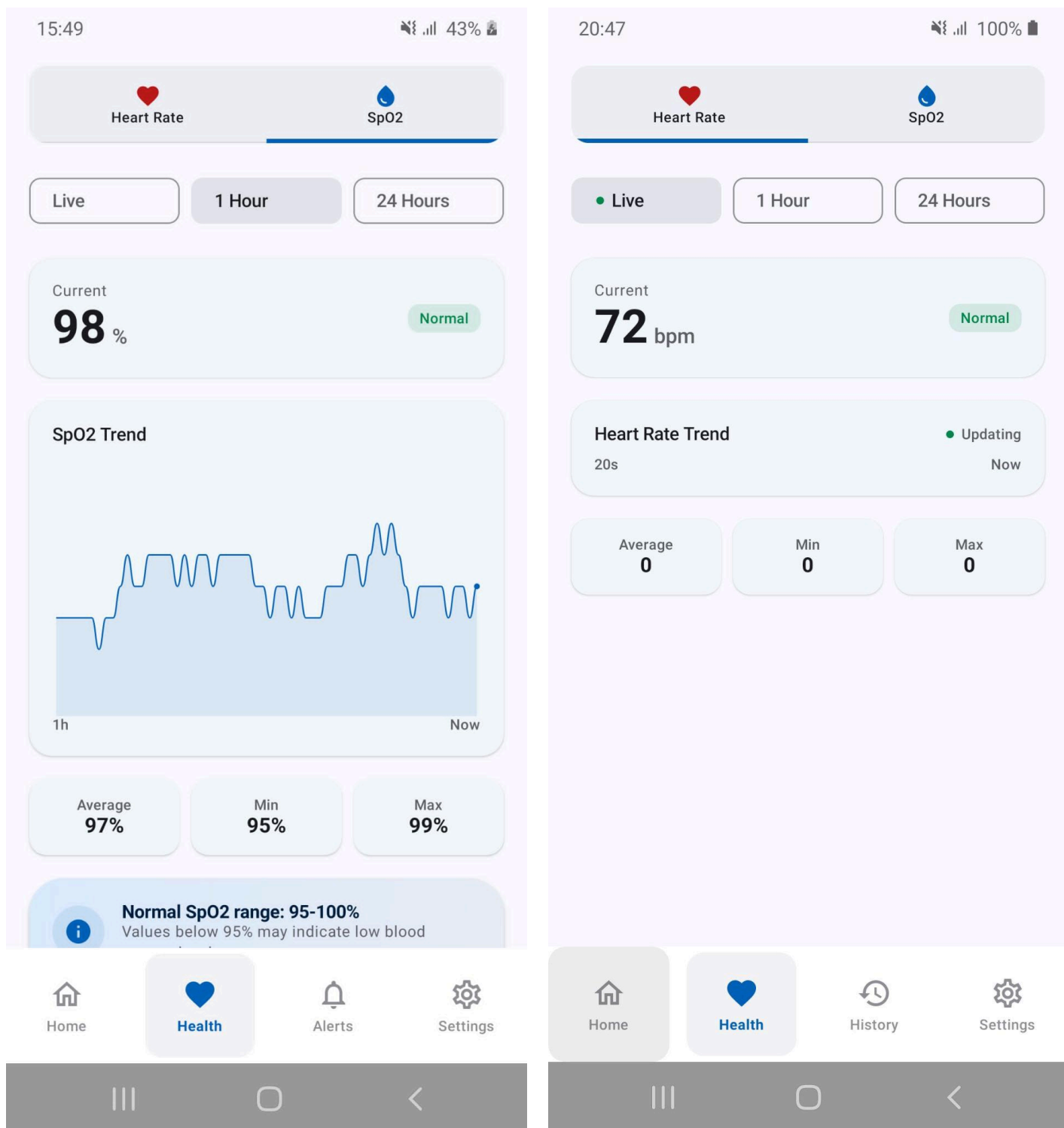


Figure 9: SpO2 interface

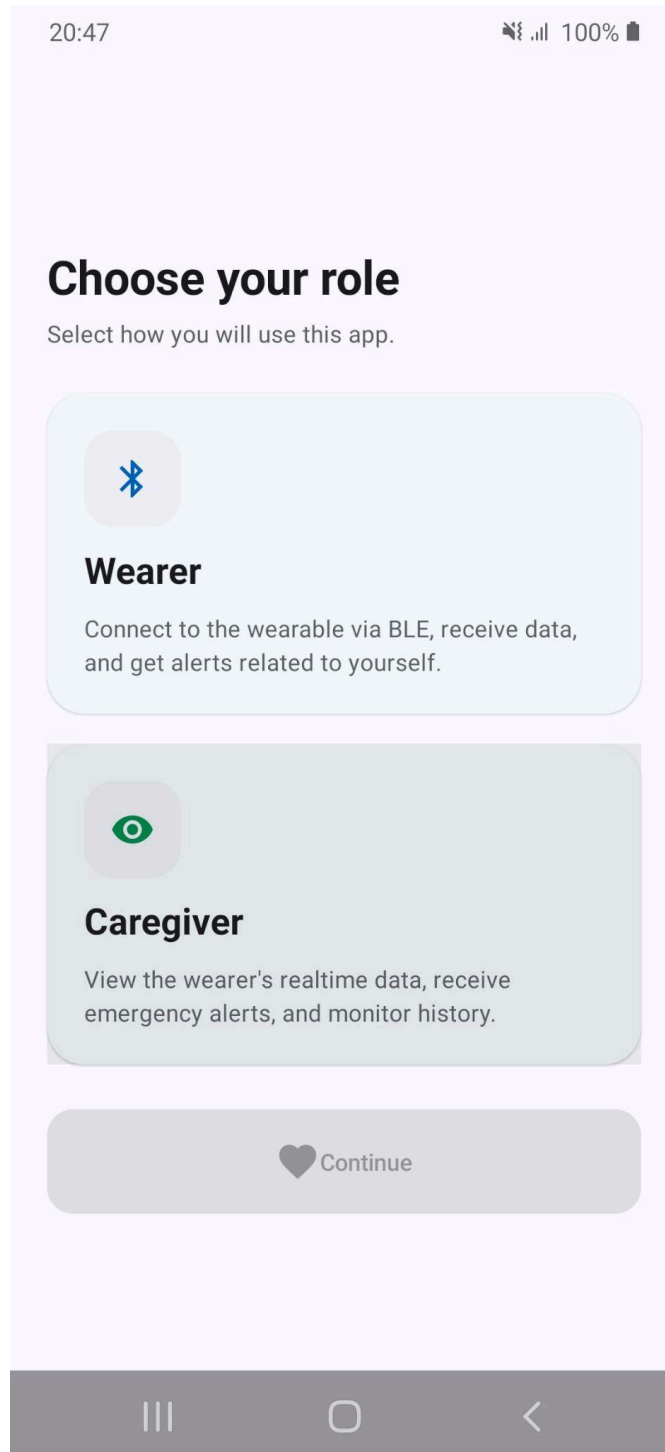


Figure 10: Bluetooth Connection Interface

1.6 Communication and Alert Module

Emergency Response Pipeline:

Fall Detected (ESP32-S3) -> BLE Notification -> Mobile App -> 15-second countdown (user can cancel) -> Automated Emergency Call/SMS -> GPS location shared (via phone)

Design rationale: (1) Edge-first processing ensures zero network latency and works without internet connectivity; (2) BLE 5.0 provides low-power, reliable short-range communication; (3) The 0.40 threshold was optimized for high recall to minimize missed falls — in safety-critical applications, occasional false alarms are preferable to missed real falls.

2. Individual Contributions

Table 5: Personal Job Assignments and Status

Task #	Task Details	Timeline	Dien	Tien	Status
1	Research elderly health issues & safety statistics	02/03-08/03	X	X	Done
2	Analyze user personas & define system requirements	09/03-15/03		X	Done
3	Benchmark commercial products & study solutions	16/03-22/03	X	X	Done
4	Design preliminary system architecture	23/03-28/03	X	X	Done
5	Acquire and preprocess datasets for AI model	29/03-10/04	X		Done
6	Design and train the core AI model	11/04-20/04	X		Done
7	Develop mobile application UI/UX and backend	29/03-20/04	X	X	In Progress

8	Optimize AI model for edge deployment	21/04-30/04	X		Done
9	Develop firmware and sensor acquisition modules	01/05-12/05	X	X	In Progress
10	Design PCB schematic and assemble SMD	13/05-25/05		X	Done
11	Integrate software, hardware, and wireless	26/05-31/05	X	X	Upcoming
12	Design the 3D enclosure	01/06-10/06		X	Upcoming
13	Evaluate system performance and reliability	10/06-20/06	X	X	Upcoming

3. Up-to-Date Results

3.1 AI Model Performance

Test set evaluation results (Balanced HR_IMU V2, threshold = 0.40):

Table 6: Results of AI Model Performance Evaluation (Test Set)

Metric	Value	Target	Status
Accuracy	90.80%	$\geq 95\%$	Close
Precision	85.41%	—	Good
Recall	98.36%	$\geq 95\%$	Excellent
F1-Score	91.43%	—	Good
False Alarm Rate	16.73%	Minimize	Needs improvement
Miss Rate	1.64%	Minimize	Excellent
ROC AUC	0.9712	—	Excellent
PR AUC	0.9669	—	Excellent

Confusion Matrix (Test Set):

Table 7: Confusion Matrix Test Set

	Predicted Non-Fall	Predicted Fall
Actual Non-Fall	204 (TN)	41 (FP)
Actual Fall	4 (FN)	240 (TP)

Key insight: Only 4 out of 244 falls were missed (1.64% miss rate), which is critical for a life-safety application. While overall accuracy (90.8%) is slightly below the 95% target, the recall of 98.36% is exceptional. The accuracy gap is primarily driven by 41 false positives, which will be addressed through model refinement and temporal filtering.

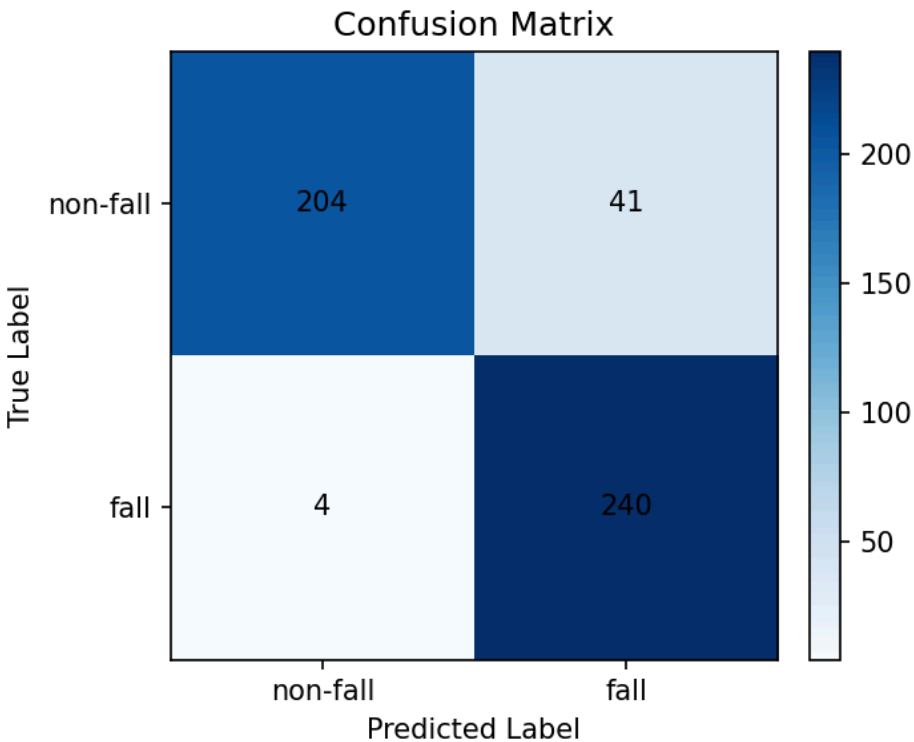


Figure 11: Confusion Matrix Heatmap

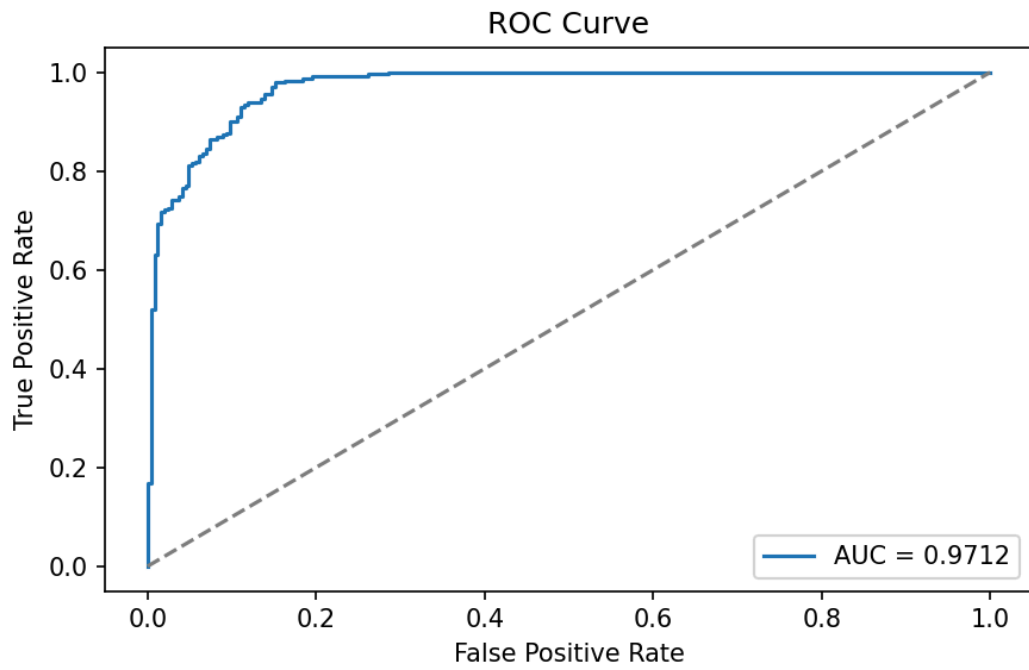


Figure 12: ROC Curve

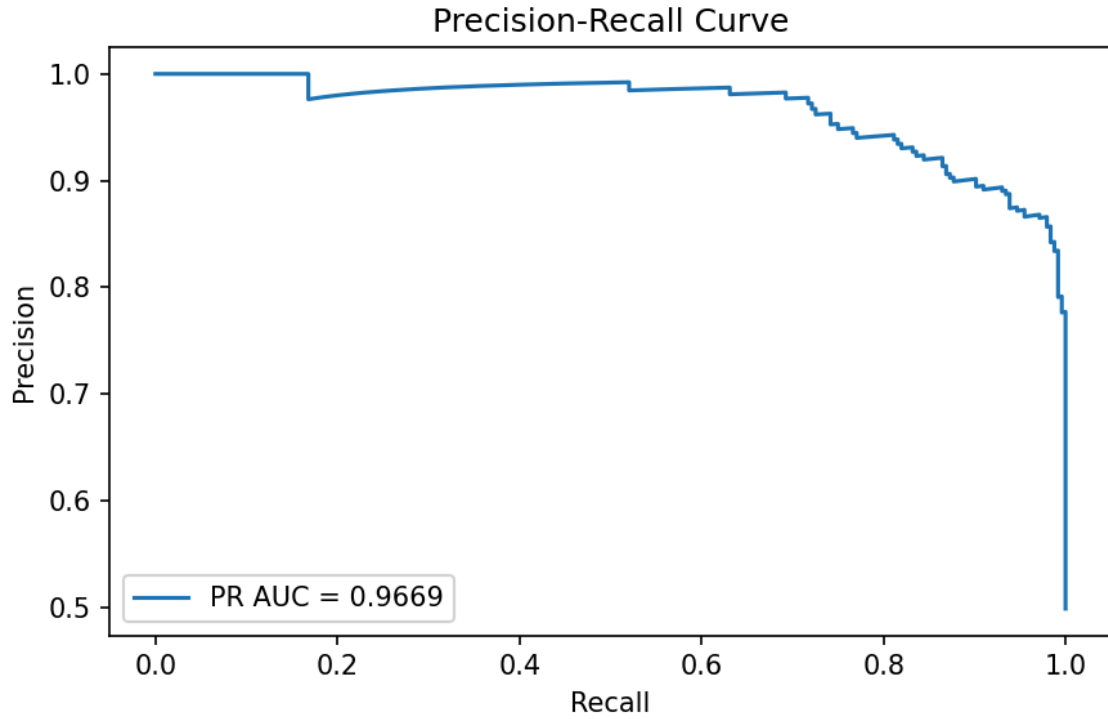
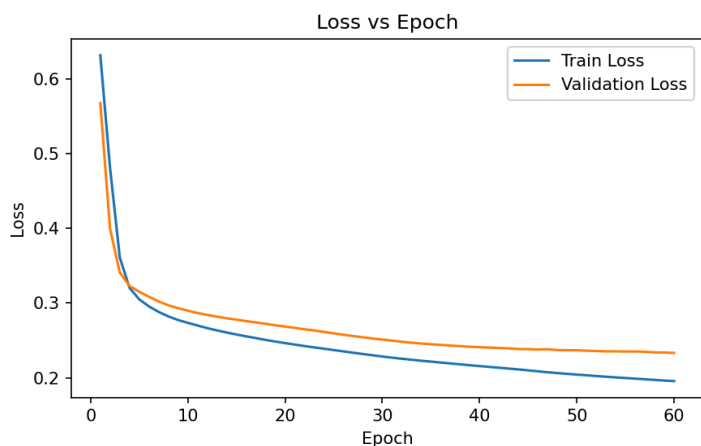
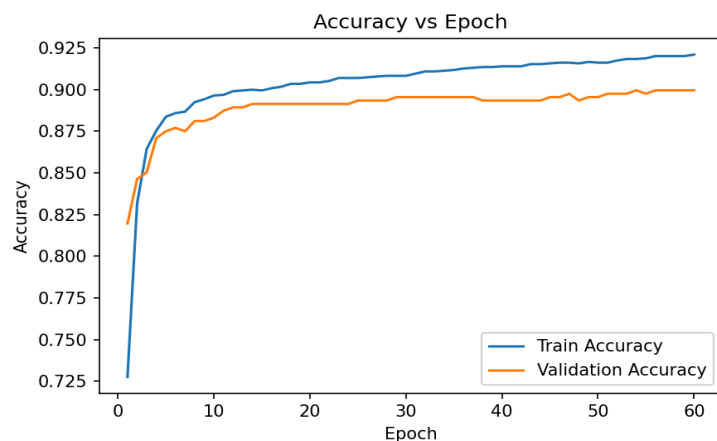


Figure 13: Precision-Recall Curve



(a): Loss and Epoch



(b): Loss and Epoch

Figure 14: Training Loss and Accuracy vs Epoch

3.2 Edge Deployment Readiness

Table 8: Deployment Readiness Indicators on Edge Devices

Deployment Metric	Value	Requirement
TFLite model size	10.71 KB	≤ 500 KB — PASS
Input shape	[1, 100, 6]	—
Output shape	[1, 1]	—
Quantization	Full INT8	ESP32-S3 optimized — PASS
Inference threshold	0.40	Tuned for max recall — PASS

3.3 Mobile Application Progress

Week 1 deliverables: Basic app framework, BLE connection setup, initial UI layout, first APK build (~46 MB).

Week 2 deliverables: Enhanced UI/UX with health data visualization, historical data viewing, improved BLE stability, updated APK (~99 MB).

Current feature status:

- BLE 5.0 device connection & pairing — Done
- Real-time HR & SpO2 data display — Done
- Health data history & trends — Done
- Fall alert notification system — In Progress
- Emergency contact auto-call/SMS — In Progress

3.4 Hardware Procurement

All critical hardware components have been identified and procurement is underway. The ESP32-S3 development board, BMI160 IMU sensor, and MAX30102 PPG sensor have been acquired. The Bill of Materials is being tracked to stay within the 2,000,000 VND budget target.

3.5 PCB Design and Assembly

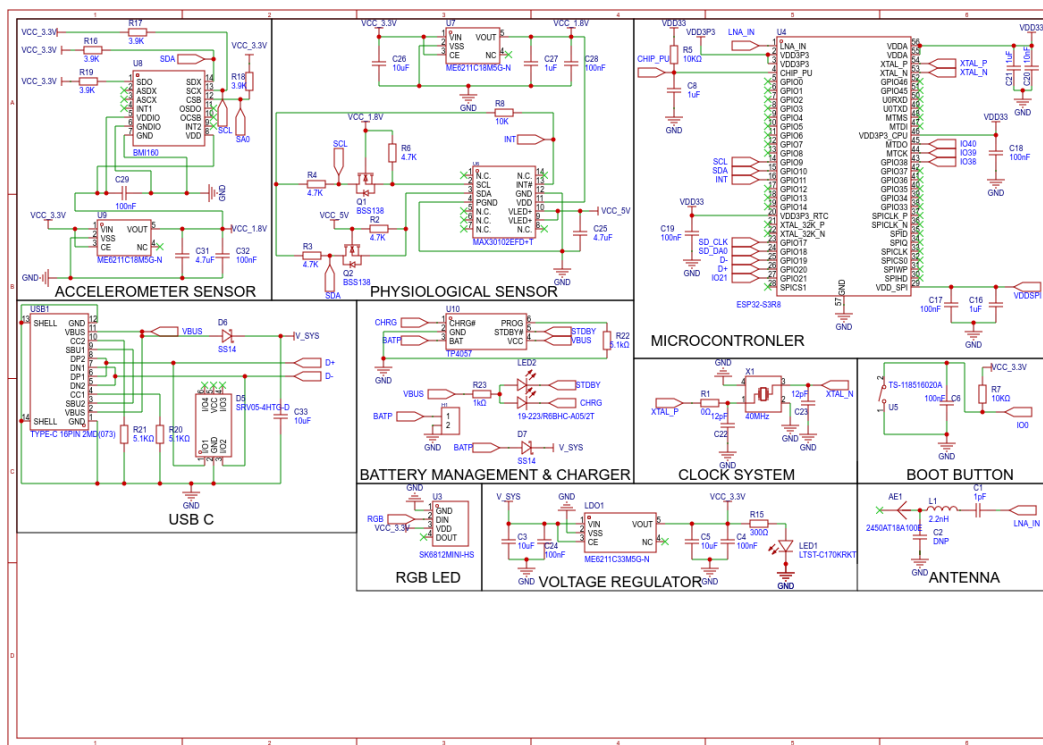


Figure 15: PCB schematic

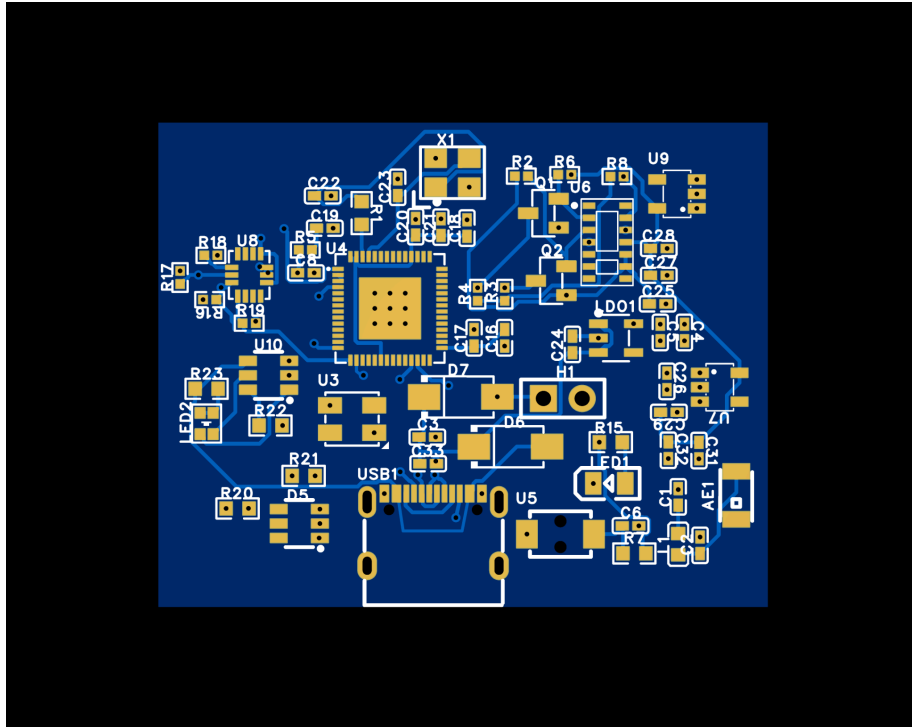


Figure 16: 2D PCB

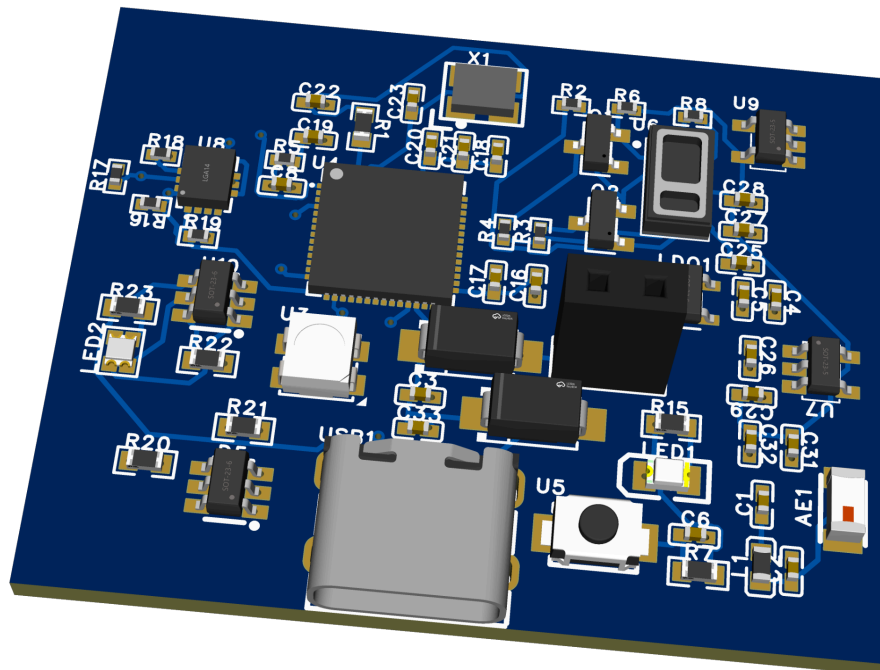


Figure 17: 3D PCB

4. Future Work

4.1 Remaining Tasks (M2 to Final)

Table 9: Remaining Work Plan (Phase M2 to Final)

Phase	Tasks	Timeline	Owner
Firmware Dev	Integrate sensor drivers (BMI160+MAX30102), deploy TFLite model	01/05-25/05	Dien + Tien
Integration	Integrate firmware + hardware + BLE + mobile app	26/05-31/05	Dien + Tien
Enclosure	Design 3D-printed wristband enclosure (Optimization)	01/06-10/06	Tien
Evaluation	System-level testing and performance benchmarking	10/06-20/06	Dien + Tien

4.2 AI Model Improvement Plan

1. Data augmentation: Add noise injection, time warping, and magnitude scaling to increase effective dataset size.
2. Model architecture refinement: Experiment with lightweight attention mechanisms while staying within size constraints.
3. False alarm reduction: Implement post-processing requiring consecutive positive windows before triggering alert.
4. Real-world validation: Collect data from the prototype wristband and fine-tune on domain-specific data.

4.3 Target Metrics for Final Milestone

Table 10: Target Indicators for the Final Milestone

Metric	Current	Target
Accuracy	90.80%	$\geq 95\%$
Recall	98.36%	$\geq 95\%$ (maintain)
False Alarm Rate	16.73%	$\leq 10\%$
Model size	10.71 KB	≤ 500 KB

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